

Demystifying dlna

The chances that you own a DLNA-capable device are high, but the chances that you are using the technology fully are pretty low. This article will help you make the most out of DLNA at home.



Have you noticed the logo near the title on products you bought recently? The latest home electronics devices come with a media-sharing technology called DLNA, which is mainly for non-technical end-users and lets them share pictures, videos and music across devices, without hassle. There are already half a billion DLNA-certified devices in the world, but awareness and use by consumers is still very low.

A few years ago, to watch videos from your PC on the TV's larger screen, you had to burn a CD/DVD and play it on the DVD player. Then devices like USB pen drives made CDs and DVDs obsolete, also overcoming the disc size limits, if you had a player with a USB port. Later, when media devices came out with LAN or Wi-Fi features, the configuration was tedious—it included configuring the IP address, starting a server on your PC, etc. DLNA technology makes all this redundant. If a DLNA-compliant server is running on your home network, DLNA clients (players) connected to the same network will discover it. It is that simple!

DLNA or Digital Living Network Alliance is an initiative for standardisation by the electronics industry leader, Sony, which began in 2003. It now has around 220 member companies, which include HP, Cisco, Broadcom, Google and

Samsung. DLNA organises devices in different classes based on the capability of the device and the interaction it will have with another device class. For example, you have a PC with music as your media source, classified as a Digital Media Server (DMS), and you want to listen to those songs on your smartphone, which is now the target device, called Digital Media Player (DMP). One device can be in more than one class, because of the interaction criteria. For example, the same smartphone that was earlier a DMP could share videos or pictures with your new LCD/LED networked TV—thus acting as a DMS, while the TV acts as a DMP.

There are three other classes: Digital Media Renderers, Controllers and Printers. In addition, there are equivalent classes exclusively for mobile devices, which support different file formats commonly used in small-screen devices. To avoid confusion in terminology, I will restrict my focus to DMS and DMP.

DLNA architecture

DLNA is based on UPnP (Universal Plug and Play), an Internet Protocol stack service. When you search the Internet for DLNA, you are more likely to find vendor articles or videos that paint DLNA as a technology in which